# Hierarchical Projection as a Validity Layer for Multilingual ESG Promise Verification

Sheng-Xun Chen, Chun-Wei Chen, Hsin-Chen Han, Shi-Yu Huang

National Central University, Taiwan • Team_10537

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## 1 Introduction

- ESG promise verification labels each paragraph with **four linked fields**, and the output is meant for audit-oriented use.
- A record is only usable if its fields agree, yet independent classifiers can combine plausible labels into an impossible tuple.
- The official metric scores each field on its own, so an invalid tuple still earns **44.3%** of the weighted field credit.

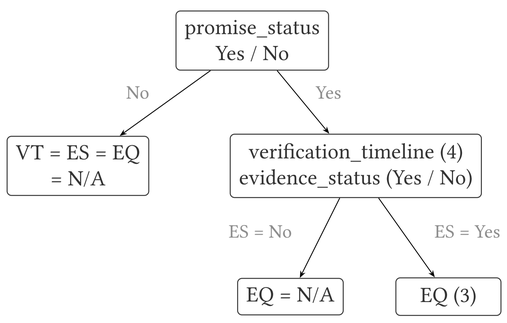
**Research question.** *Can a retraining-free decision rule guarantee legal outputs, and how does it change field-wise versus whole-tuple scores?*

## 2 Task & Data

- Fields: promise status (PS), verification timeline (VT), evidence status (ES), evidence quality (EQ).
- Hard rules: PS = No $\Rightarrow$ VT = ES = EQ = N/A; ES = No $\Rightarrow$ EQ = N/A.
- Only **17 of 120** label combinations are legal.
- **AI CUP 2026**: 2,000 Chinese paragraphs from 49 reports, all legal (15 of 17 states observed).
- **ML-Promise**: English, French, Japanese (400 paragraphs each), Korean (500 PDF pages).

## 3 Methodology

(a) Task hierarchy




$1 + 4 \times (1 + 3) = 17$ legal tuples

of $2 \times 5 \times 3 \times 4 = 120$ combinations

*the other 103 are invalid*

(b) Decision rules: output rule × calibration

base probabilities (identical for every method)

*calibration*

| | none | global | conditional |
|---|---|---|---|
| independent argmax *may be invalid* | **M0** | — | — |
| projection *always valid* | **M1** | **M2** | **M3** |
| 17-state decoding *always valid* | **M4** | **M5** | **M6** |

*the seven rules are alternatives, not stages: every cell consumes the same base probabilities*

Figure 1: the label hierarchy and the seven alternative decision rules, all reading the same base probabilities

### Validity-preserving output rules

- **Projection**: greedy and top-down. Pick PS; under PS = Yes take the best substantive VT and ES; under ES = Yes the best substantive EQ. **Always legal**, no retraining.
- **17-state decoding**: return the legal state with the highest summed log-score.
- Calibration: none, global, or conditional class biases.

### Evaluation

- `hfl/chinese-roberta-wwm-ext-large`, four independent heads.
- Five-way cross-fitting, three seeds, document-disjoint split.
- Paired report-cluster bootstrap, Holm-adjusted.
- External: 4 backbones × $\lambda \in \{0, 0.3\}$ in each of 4 languages = **32 arms**.

## 4 Results: Chinese Anchor

| ID | Calibration | Decoding | PS | VT | ES | EQ | wF1 (official) | Tuple acc. | Invalid % |
|---|---|---|---|---|---|---|---|---|---|
| M0 | None | Independent | 0.796 | 0.464 | 0.650 | 0.424 | 0.572±0.003 | 0.359±0.007 | 12.55 |
| M1 | None | Projection | 0.796 | 0.467 | 0.651 | 0.418 | 0.571±0.003 | 0.394±0.004 | 0.00 |
| M2 | Global | Projection | 0.796 | 0.493 | 0.650 | 0.418 | 0.574±0.004 | 0.428±0.004 | 0.00 |
| M3 | Conditional | Projection | 0.796 | 0.501 | 0.651 | 0.416 | 0.575±0.003 | 0.432±0.004 | 0.00 |
| M4 | None | 17-state | 0.799 | 0.468 | 0.648 | 0.420 | 0.571±0.005 | 0.388±0.003 | 0.00 |
| M5 | Global | 17-state | 0.796 | 0.493 | 0.649 | 0.423 | 0.576±0.001 | 0.430±0.001 | 0.00 |
| M6 | Conditional | 17-state | 0.784 | 0.494 | 0.634 | 0.418 | 0.567±0.008 | 0.427±0.005 | 0.00 |

Table 2: three-seed means ± s.d., document-disjoint

| Metric | Contrast | Δ | 95% CI | $p_{\text{Holm}}$ |
|---|---|---|---|---|
| wF1 (official) | M1-M0 | -0.001 | [-0.006, 0.003] | 1.000 |
| | M3-M2 | +0.001 | [-0.004, 0.006] | 1.000 |
| | M4-M1 | +0.000 | [-0.005, 0.005] | 1.000 |
| | M6-M3 | -0.008 | [-0.018, 0.002] | 0.416 |
| | M6-M5 | -0.008 | [-0.016, -0.001] | 0.171 |
| Tuple accuracy | M1-M0 | +0.035 | [0.028, 0.043] | **0.001** |
| | M3-M2 | +0.004 | [-0.001, 0.008] | 0.370 |
| | M4-M1 | -0.006 | [-0.010, -0.002] | **0.028** |
| | M6-M3 | -0.005 | [-0.015, 0.004] | 0.505 |
| | M6-M5 | -0.004 | [-0.011, 0.004] | 0.505 |

Table 4: pre-specified paired contrasts, 95% intervals

- Projection removes every invalid tuple (**12.55% → 0**) and raises tuple accuracy by **+0.035** ($p_{\text{Holm}} = .001$), while official wF1 does not move ($\Delta = -0.001$, $p_{\text{Holm}} = 1.000$).
- Searching all 17 states is not better: joint decoding loses **−0.006** tuple accuracy to projection ($p_{\text{Holm}} = .028$).

## 5 Multilingual Replication

| Corpus | Rows | M0 invalid % | $N/A$ net | Subst. net | Δ wF1 | Δ tuple acc. |
|---|---|---|---|---|---|---|
| Chinese (AI CUP) | 2,000 | 12.55 | +395 | -264 | -0.0011 | **+0.0350** |
| English | 400 | 23.17 | +148 | -86 | +0.0032 | **+0.0725** |
| French | 400 | 21.08 | +150 | -107 | +0.0036 | **+0.0625** |
| Japanese | 400 | 31.25 | +183 | -245 | -0.0131 | **+0.0708** |
| Korean | 500 | 22.73 | +103 | -148 | -0.0192 | **+0.0293** |

Table 7: projection (M1) vs. argmax (M0), one preselected arm per corpus

- Tuple accuracy rises in **32/32** external arms; wF1 in only **14/32** (EN 7/8, FR 5/8, JA 0/8, KO 2/8).
- These are within-arm directions, not a ranking of languages.

## 6 Why the Metrics Disagree

- A wrong parent makes projection overwrite a correct child with N/A; a right parent lets it repair an unsupported child.
- N/A credit goes up and substantive credit goes down in all five corpora, yet whole tuples improve in every one.
- Joint decoding revises PS on **339 of 12,000** rows: 225 become correct and 114 wrong, but whole-row correctness on them falls **114 → 47**.
- Fixing a parent rarely saves a row whose children stay wrong; breaking a correct parent always loses it.

## 7 Implications for Audit Use

- Validity is a **precondition** for a usable record, not proof that the record is right.
- A repair can hide an inconsistency, so a deployed validity layer should **log the rows it repaired**.
- Pick the metric by how outputs are used:
  - records: tuple accuracy among valid outputs
  - single-field actions: that field's F1, or rare-class recall
  - portfolio statistics: class-level scores, keeping validity
  - leaderboards: official wF1, but not for gaps under 0.008

## 8 Conclusion

- Hierarchical projection is a **retraining-free validity layer**: it eliminates invalid tuples by construction.
- It improves whole-tuple accuracy on the Chinese anchor and in all 32 external arms.
- Its field-score effect varies across models and corpora; its logical guarantee does not.

**Takeaway.** For audit use, make validity a precondition, and pick the metric that matches the unit at which errors have consequences.

**References**

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**Code & reproduction**

github.com/tom1030507/ntcir19-esg-validity-layer